

From 2D Keypoints to 3D Bodies: A Literature Review on Human Pose Estimation

Yiqiao Liu

Taijia Liang

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1 Introduction

[Placeholder] Motivate the topic, define the research question, and preview the three landmark methods reviewed below.

2 Background

[Placeholder] Briefly review heatmap-based 2D pose lineage (CPM, Stacked Hourglass) and the SMPL parametric body model that underlies HMR.

3 Three Core Methods

3.1 OpenPose: Real-time Multi-person 2D Pose via PAFs

The challenges of real-time multi-person 2D estimation lie in the significant uncertainty caused by the unknown number of people in the image, spatial interference due to interactions and overlaps between people, and the substantial increase in spatial complexity as the number of people increases. Cao et al.'s Part Affinity Fields (PAFs) [1] effectively address these issues. Compared to traditional top-down methods (reusing single-person pose assessment techniques) and traditional bottom-up techniques, they achieve efficient identification and reconstruction of unstructured pairwise relationships for a variable number of human body parts through PAFs (a set of 2D vector fields) and Confidence Maps (a set of confidence maps of human body parts) [1].

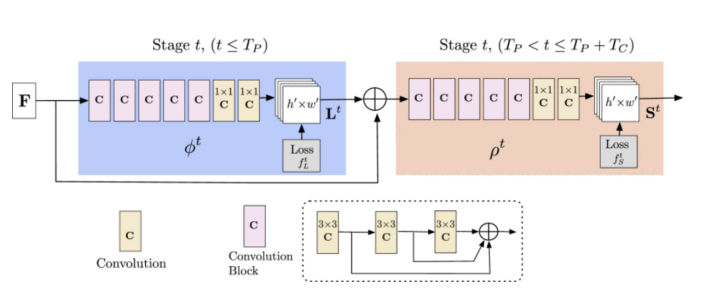


Figure 1: Multi-stage prediction network used in OpenPose.

As shown in Figure 1, a multi-stage prediction network is used for convolutional neural networks (CNNs) [1]. Based on refining PAF (the degree of correlation between individual human limbs), Gaussian peak convolution is used to further convolve the confidence map (the location of individual human parts) [1]. By changing the 7×7 convolutional kernel to three consecutive 3×3 convolutional kernels, the operation count is effectively reduced from $2 \times 7^2 - 1 = 97$ to 51 [1]. A spatially weighted loss handles unlabelled individuals. For multi-person parsing, Cao et al. simplify the bipartite matching to a spanning-tree skeleton solved by greedy inference [1].

3.2 HRNet: Deep High-Resolution Representation Learning

[Placeholder] Cover (i) the parallel multi-resolution architecture and why preserving high-resolution features beats encoder-decoder recoveries, (ii) the repeated multi-scale fusion mechanism, (iii) COCO/MPII SOTA numbers, (iv) top-down dependency on an external detector.

3.3 HMR: End-to-end Recovery of Human Shape and Pose

[Placeholder] Describe (i) the SMPL parameter regression target and iterative error feedback, (ii) the adversarial body prior that sidesteps 3D labels, (iii) 2D reprojection plus 3D losses when available, (iv) Human3.6M results and the now-canonical line of successors (PyMAF, HMR 2.0).

4 Connections, Differences, and Limitations

[Placeholder] Synthesise the three methods with one comparison table and a short narrative on how they compose into a $2D \rightarrow 3D$ pipeline.

5 Future Directions and Conclusion

[Placeholder] Summarise the $2D \rightarrow 3D$ progression and outline three future directions: foundation models (Sapiens), video extensions (VIBE), and SMPL + 3DGS animatable avatars.

References

- [1] Z. Cao, G. Hidalgo, T. Simon, S.-E. Wei, and Y. Sheikh, “OpenPose: Realtime multi-person 2D pose estimation using part affinity fields,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 43, no. 1, pp. 172–186, 2019.
- [2] K. Sun, B. Xiao, D. Liu, and J. Wang, “Deep high-resolution representation learning for human pose estimation,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 5693–5703.
- [3] A. Kanazawa, M. J. Black, D. W. Jacobs, and J. Malik, “End-to-end recovery of human shape and pose,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 7122–7131.